

MAX YUAN

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EDUCATION

University of Waterloo | Mechatronics Engineering, Honours Co-op | Sept 2026 - Sept 2031

TECHNICAL SKILLS

Languages: Java, Python

Skills: Robotics Simulation, Reinforcement Learning (PPO), PID control, Odometry, Inverse Kinematics

Libraries/Tools: Isaac Sim, Isaac Lab, rsl_rl, ROS2, Rviz, Onshape, OpenCV, YOLOv8

Strengths: Proactive learner, Self-driven, Passionate, Feedback-driven

PROJECTS

Custom Quadruped Locomotion — Unitree Go2 PPO in Isaac Lab | Aug 2026 – Sept 2026

Solo project: custom ManagerBasedRLEnv for flat-terrain Go2 locomotion, trained with PPO via NVIDIA Isaac Lab

- Custom built RL env: 48-dim observation space, 12-DOF joint-position actions, and custom reward stack
- Resolved an agent crawling gait issue by implementing a custom narrow Gaussian reward kernel, tracking height error to reward maintaining an upright posture
- Implemented a sigmoid curriculum that ramps the foot-lift reward based on velocity-tracking performance, forcing the robot to master balance before refining its gait
- Repo: github.com/NotMax08/unitree_go2_ppo_isaacLab

Autonomous Maze Navigation — Nova Carter in Isaac Sim | May 2026 – July 2026

Solo project: full ROS2 Nav2 stack with a Nova Carter differential-drive robot via NVIDIA Isaac Sim 5.1 on Ubuntu

- Built end-to-end autonomous navigation: SLAM mapping with slam_toolbox, AMCL localization, Nav2 planning with Regulated Pure Pursuit controller, ROS2 as a middleware framework and Rviz for debugging
- Constructed the full TF tree (map → odom → chassis_link → sensors) from scratch via Isaac Sim action graphs; debugged QoS mismatches, sensor rate bottlenecks, transformation frame authority conflicts
- Diagnosed inconsistent warped SLAM map through topic loop rates caused by scan misalignment during turns
- Repo: github.com/NotMax08/isaac-sim-maze-slam-nav2

RICO — Voice-Controlled Tool-Grabbing Robotic Arm (Eureka Hacks) | May 2026

- Built a 4-DOF robotic arm utilizing inverse kinematics that retrieves user-requested tools from natural-language voice commands, integrating Groq LLM to parse speech into executable arm actions
- Applied a homography transformation matrix with OpenCV to convert an angled webcam feed into a bird's-eye view, improving tool-localization accuracy; fine-tuned a YOLOv8 tool-detection model on Roboflow datasets

Co-Captain & Lead Programmer — FIRST Tech Challenge Robotics Team #18844 | Sept 2023 – Apr 2025

2025 FIRST World Championships — 1st Connect Award Winner / Ochoa Division

- Point-to-Point PID movement system: tuned gains for 3 positional PID controllers (x, y, heading) enabling autonomous pathing with minimal error
- Implemented 2-wheel odometry + IMU for robot localization, converting local pose into global coordinates
- Automated sample pickup system with real-time object detection using OpenCV and a custom-trained CNN for distance and sample orientation

LEADERSHIP EXPERIENCE

Co-Captain & Outreach Coordinator — FIRST Tech Challenge Robotics Team #18844 | Sept 2023 – Apr 2025

- Co-led Team 18844 PrOteens as Team Canada representatives on an international stage, overseeing programming systems integration and technical decisions
- Developed project-based LEGO SPIKE Prime lessons for neurodiverse and beginner students from under-resourced background